Family Finance Tracker

Product Requirements Document

Version 1.0
March 2026
Selva & Akshaya

1. Product Overview

Family Finance Tracker is a personal web application designed for Selva (Nvidia) and Akshaya (Microsoft) to replace their manual Google Sheets-based financial tracking system. The application will provide a structured, year-wise (Indian Financial Year: April–March) view of combined and individual income, investments, and expenses. It will serve as the single source of truth for the family's financial picture, with live stock price and currency data, and will grow into a full financial planning dashboard over time.

Attribute	Detail
Product Name	Family Finance Tracker
Financial Year	Indian FY (April 1 — March 31)
Primary Users	Selva (Nvidia) and Akshaya (Microsoft)
Platform	Web Application (mobile-first, desktop-compatible)
Version Scope	v1.0 — Income Tracking + US RSU Tab
Future Scope	Investments, Expenses, Portfolio, Multi-year Graphs
Data Source	Manual user entry + live API (stock prices, forex)

2. Goals & Objectives

- Replace manual Google Sheets tracking with a structured, error-proof web interface.
- Track all income components (fixed salary, RSU, ESPP, EPF, bonuses) for both Selva and Akshaya individually and combined, per Indian Financial Year.

- Handle the variable nature of income: tax-adjusted take-home, quarterly RSU vesting, irregular bonuses.
- Track US-listed RSU vesting events for both MSFT and NVDA with live USD→INR conversion and real-time stock prices.
- Provide a foundation for multi-year income growth analysis (charts, YoY comparison).
- Reserve tabs for future Investment and Expense tracking modules.

3. Scope

3.1 In Scope — v1.0

- Income Tracking Tab (core module)
- US RSU Vesting Tracker Tab
- Live data: USD/INR exchange rate, MSFT share price, NVDA share price
- Indian Financial Year (April–March) switching
- Combined + individual (Selva / Akshaya) income views
- Yearly summary with monthly breakdown

3.2 Out of Scope — v1.0 (Planned for Future)

- Investment Tracking Tab (EPF corpus, RSU portfolio, ESPP, Debt funds, Baby Education Fund)
- Expense Tracking Tab
- Portfolio Tracker (current market value of all holdings)
- Multi-year projection / scenario modelling
- User authentication / multi-user access
- Mobile native app

4. Users & Context

There are two users, both acting as administrators of their own data:

User	Employer	Income Components
Selva	Microsoft	Monthly Take-Home (post-tax), EPF (fixed), ESPP (auto-deduction), RSU (quarterly vesting in USD), Cash Bonus, Joining Bonus, Car Lease Component
Akshaya	Nvidia	Monthly Take-Home (post-tax), EPF (fixed), ESPP (auto-deduction), RSU (quarterly vesting in USD), Cash Refund

5. Application Structure & Tabs

The application is structured as a tabbed single-page web app. Each tab is a self-contained module. The tab bar is always visible.

Tab	Description	v1.0 Status
Income	Track all monthly and periodic income for Selva and Akshaya	Active
US RSU Tracker	Log and value MSFT/NVDA RSU vesting events in USD and INR	Active
Investments	Track EPF, ESPP, RSU portfolio, debt funds, baby education fund	Placeholder
Expenses	Track monthly expenses vs budget across categories	Placeholder
Portfolio	Live valuation of all current holdings	Placeholder

6. Income Tracking Tab — Functional Specifications

6.1 Financial Year Selector

- The app tracks data per Indian Financial Year (FY), running April 1 to March 31.
- A prominent FY selector (e.g., dropdown: “FY 2025–26”) is visible at the top of the page.
- Switching FY reloads all tabs with data for that year.
- On first launch, the current FY is pre-selected.

6.2 Income Components Tracked

The following components are tracked per person, per month within the FY:

Component	Person	Frequency	Notes
Take-Home Salary	Both	Monthly	Post-tax, post-EPF, post-ESPP. Variable each month due to tax slab changes and variable deductions.
EPF (Employee)	Both	Monthly	Fixed monthly amount derived from base pay. Rarely changes; set once per FY, can be edited.
ESPP Contribution	Both	Monthly (auto-deducted)	Fixed % of base salary, deducted before take-home. Logged as an outflow from gross.
RSU Vesting	Both	Quarterly (variable)	Number of units vested and vesting month vary by quarter. USD value × stock price × USD/INR at vest date.

Cash Bonus	Selva	Annual/Ad-hoc	E.g., Microsoft annual cash bonus, joining bonus.
Car Lease Component	Selva	Monthly	Microsoft provides a car lease component added to CTC; tracked separately.
Cash Refund	Akshaya	Ad-hoc	Occasional cash refunds from Nvidia (e.g., expense reimbursements).

6.3 Data Entry — Monthly Salary

- User selects a month (April–March) and a person (Selva / Akshaya).
- Fields available: Take-Home amount (INR), EPF Employee Contribution (INR, pre-filled from last entry), ESPP Contribution (INR, pre-filled), Car Lease Component (Selva only), and any ad-hoc cash items (labelled freeform).
- EPF and ESPP values are sticky: once set for the FY, they auto-populate subsequent months unless explicitly changed (EPF changes when base salary changes).
- The system shows the calculated “Gross Approximate” = Take-Home + EPF + ESPP + Tax withheld (if entered separately) as a reference.

6.4 Data Entry — RSU Vesting

- RSU vesting is entered separately per event (not per month grid).
- Fields: Person (Selva/Akshaya), Vest Month, Number of Units Vested, Stock (MSFT or NVDA), USD Price at Vest, USD/INR Rate at Vest, Calculated INR Value (auto-computed).
- Stock price and USD/INR at vest date can be manually overridden or auto-fetched if the vest date is today.
- Multiple RSU vesting events can occur in the same quarter; each is a separate entry.
- The vested INR amount is reflected in the monthly income row for the vest month.

6.5 Income Views

- Combined View: Shows merged monthly income table for both Selva and Akshaya as a household.
- Individual View: Toggle to see Selva-only or Akshaya-only income.
- Monthly Breakdown Table: Rows = income component, Columns = April to March. Last column = FY Total.
- FY Summary Cards: Total Take-Home, Total RSU, Total EPF, Total ESPP, Grand Total Income (INR).
- All INR values rounded to nearest rupee. USD values shown in original currency with INR equivalent.

6.6 Live Market Data

- A header bar or info strip shows: Current MSFT Price (USD), Current NVDA Price (USD), Current USD/INR Rate.

- Data is fetched from a public finance API (e.g., Yahoo Finance, Alpha Vantage, or exchangerate.host).
- Refresh button / auto-refresh every 15 minutes when the tab is active.
- If data fetch fails, the last cached value is shown with a timestamp and a warning indicator.
- Live prices are used as default when entering new RSU events; user can still override.

7. US RSU Vesting Tracker Tab — Functional Specifications

7.1 Purpose

This tab provides a dedicated register of all MSFT and NVDA RSU vesting events across financial years. It serves as an audit trail for RSU income, useful for tax filings and tracking equity compensation growth.

7.2 Data Fields Per Vesting Entry

Field	Type	Notes
Person	Dropdown	Selva / Akshaya
Company / Stock	Dropdown	MSFT / NVDA
Vest Date	Date Picker	Actual vesting date (dd/mm/yyyy)
Financial Year	Auto-computed	Derived from vest date
Units Vested	Number	Number of RSU shares vested
Stock Price at Vest (USD)	Number	Can be auto-fetched or manually entered
USD/INR at Vest	Number	Exchange rate on vest date; auto-fetch or manual
Gross USD Value	Auto-computed	Units × Stock Price
Gross INR Value	Auto-computed	Gross USD × USD/INR Rate
Tax Withheld Units	Number (optional)	Units withheld by employer for TDS
Net Units Received	Auto-computed	Units Vested − Tax Withheld Units
Net INR Value	Auto-computed	Net Units × Stock Price × USD/INR
Notes	Text	E.g., grant ID, grant type (refresher / sign-on)

7.3 RSU Summary View

- Filter by: FY, Person, Stock.
- Total units vested, total USD value, total INR value — per FY and all-time.
- Current portfolio value card: units held × current live price (links to portfolio tab in future).

- FY-wise RSU table showing vesting schedule (past and upcoming) — upcoming rows derived from grant schedule entered by user.

7.4 Upcoming RSU Schedule (Optional Input)

- User can enter future RSU grants: Grant Date, Cliff, Vesting Schedule (e.g., quarterly over 4 years).
- System generates projected vest dates and unit counts.
- Projected rows are shown in muted style (not counted in actuals until manually confirmed).

8. Future Tabs — Placeholder Specifications

The following tabs will be built in future versions. They are included in v1.0 as disabled/placeholder tabs so the navigation structure is set from day one.

8.1 Investments Tab (v2.0)

- Track: EPF corpus (Selva + Akshaya), ESPP purchases and current value, Debt Funds (Edelweiss Equity Savings, PPDAAF), Baby Education Fund, PPF.
- Monthly investment entries by category.
- Target vs Actual tracking (Debt Target: 18L, Equity Target: TBD).
- Live fund NAV or estimated values where applicable.

8.2 Expenses Tab (v2.0)

- Monthly budget vs actuals across fixed categories: Rent, Groceries, Maid & Utilities, Shopping, Parents, Gifts, Travel (Vacation / Non-vacation), Eating Out, Weekend Fun, Medical, Baby, Car Loan, Miscellaneous, and ad-hoc categories.
- Recurring vs Non-Recurring split (matching current sheet structure).
- Per-expense notes / footnotes (matching current annotation style).
- Budget Left indicator with colour coding (green/amber/red).

8.3 Portfolio Tab (v3.0)

- Live valuation of MSFT and NVDA shares held (vested units minus sold units).
- EPF balance tracker.
- Debt fund balances.
- Total corpus view: Equity + Debt + EPF/PPF broken down vs targets.

9. Multi-Year Analytics (v2.0+)

Once data exists for 2+ financial years, the following analytical views become available:

- Income Growth Chart: Year-over-year line/bar chart of total income (combined and individual) broken down by component (Salary, RSU, ESPP, Bonus).
- Savings Rate Trend: (Total Income − Total Expenses) / Total Income per FY.
- RSU Appreciation: Value of vested RSUs at vest date vs current market value.
- Projection Mode: Input expected salary hike %, RSU grant schedule, and project future income for 3–5 years.
- All charts are interactive (hover tooltips, filter by person or component).

10. Non-Functional Requirements

10.1 Performance

- Page load: < 2 seconds on a standard mobile connection.
- Live data fetch: < 3 seconds. Cached fallback shown immediately; live data updates in background.
- All interactions (tab switching, data entry, calculations) respond in < 200ms.

10.2 Usability

- Mobile-first design: all tables scrollable horizontally on small screens. Month columns collapse to abbreviations (Apr, May …).
- Accessible on latest Chrome, Safari, Firefox (desktop and mobile).
- Keyboard-accessible data entry forms.
- Numeric fields: INR values formatted with Indian number formatting (e.g., ₹12,57,305).

10.3 Data Persistence

- All user-entered data is persisted in a backend database (Postgres or SQLite via a lightweight API).
- Alternatively for v1.0: localStorage with JSON export/import as a fallback if no backend is set up.
- Data export: Full FY data exportable as CSV or Excel at any time.
- No data loss on browser refresh or tab switch.

10.4 Security & Privacy

- This is a single-household private app. No multi-tenancy required in v1.0.
- If hosted publicly, protect with a simple password or HTTP Basic Auth.
- API keys for stock/forex data must not be exposed in client-side code (proxy via backend).

10.5 Technology Preferences

- Frontend: React (SPA) with Tailwind CSS. Recharts or Chart.js for visualisations.
- Backend (optional for v1.0): Node.js/Express or Python/FastAPI with SQLite.
- Live data: Free-tier finance API (Yahoo Finance unofficial, or Alpha Vantage free tier, or exchangerate.host for forex).
- Deployment: Vercel / Netlify (frontend), Render / Railway (backend if needed).

11. Data Model (v1.0)

The following entities capture all income data for v1.0:

MonthlyIncome

Field	Type	Description
id	UUID	Primary key
person	Enum	SELVA | AKSHAYA
financial_year	String	e.g., “FY2025-26”
month	Enum	APR … MAR (1–12 mapped to Apr=1)
take_home_inr	Number	Post-tax, post-deduction salary
epf_employee_inr	Number	Employee EPF contribution
espp_contribution_inr	Number	ESPP amount auto-deducted
car_lease_inr	Number	Selva only; 0 for Akshaya
ad_hoc_items	JSON Array	[{label, amount_inr}] for bonuses, refunds, etc.
notes	Text	Free-form notes for the month

RsuVestingEvent

Field	Type	Description
id	UUID	Primary key
person	Enum	SELVA | AKSHAYA
stock	Enum	MSFT | NVDA
vest_date	Date	Actual vesting date
units_vested	Number	Gross units vested
tax_withheld_units	Number	Units withheld for TDS (0 if not applicable)
stock_price_usd	Number	Share price on vest date in USD
usd_inr_rate	Number	Exchange rate on vest date
gross_usd_value	Computed	units_vested × stock_price_usd

gross_inr_value	Computed	gross_usd_value × usd_inr_rate
net_inr_value	Computed	(units_vested − tax_withheld_units) × stock_price_usd × usd_inr_rate
grant_id	String (optional)	For reference / audit
notes	Text	Free-form notes

12. Open Questions & Decisions

#	Question	Decision
1	Should the app have a login/auth layer in v1.0?	TBD — Likely HTTP Basic Auth for simplicity if hosted remotely
2	Backend or purely client-side storage (localStorage)?	TBD — localStorage + JSON export for v1.0; backend in v2.0
3	Which stock/forex API to use?	TBD — Yahoo Finance (free, unofficial) or Alpha Vantage (free tier, 5 calls/min)
4	Should ESPP be tracked as both a deduction and an asset?	TBD — v1.0 tracks as deduction from take-home; asset value in Investments tab
5	How to handle Selva EPF — Microsoft uses a different EPF base?	Confirmed: both EPF amounts entered manually and can differ month-to-month
6	Upcoming RSU grants — should they be modelled in v1.0 or v2.0?	TBD — Nice-to-have for v1.0 RSU tab, not blocking

13. Roadmap

Version	Features
v1.0	Income Tracking Tab (full), US RSU Vesting Tracker Tab, Live market data strip, FY selector, Combined + individual views, CSV export
v1.1	Upcoming RSU schedule / grant modelling, Income projection with salary hike %
v2.0	Investments Tab (EPF, ESPP, Debt funds, Baby Education Fund), Expenses Tab (budget vs actuals, notes), Multi-year income growth charts
v3.0	Portfolio Tab (live holding valuations), Savings rate trend, Total corpus vs target dashboard
v3.1+	Income scenario modelling (promotion, job change), Tax estimation helper

End of Document — Family Finance Tracker PRD v1.0